

Requirements

Definitions

Term	Definition
Employee	A person registered in the system who can be scheduled to shifts.
Shift	A defined work period (morning or evening)
Role	A function an employee performs during a shift
Driver	An employee holding a valid driving license who can be assigned to deliveries
Storekeeper	An employee responsible for loading and unloading goods at a branch
Site (Branch)	A physical location that serves as a source or destination in a delivery
Delivery	The transport of goods from a source site to a destination site, assigned a truck and a driver
Delivery Certificate	A document generated for a delivery, recording its details and contents
Delivery Zone	A geographic grouping of destination sites used for delivery planning
Truck	A vehicle defined by its number, net weight, maximum carrying weight, and required license type, used to perform deliveries
Shipment	The goods loaded onto a truck for a delivery, its total weight must not exceed the truck's capacity
License Type	The category of driving license required to operate a particular truck
Source Branch	The site from which a delivery originates
Destination Branch	The site to which a delivery is sent

Requirements

Id	Module	Description	Functional/N on-Functional	Priority	Risk	Iteration
1	Employee Management	The system shall allow registering new employees and storing their personal details.	Functional	MH	Low	Previous
2	Employee Management	The system shall store each employee's employment start date.	Functional	MH	Low	Previous
3	Employee Management	The system shall maintain employee availability by day and shift.	Functional	MH	Low	Previous
4	Employee Management	The system shall maintain employee roles and driver license qualifications.	Functional	MH	Low	Current
5	Shift Management	The system shall manage shifts while considering transportation-planning requirements.	Functional	MH	High	Current
6	Shift Management	The system shall support morning and evening shifts.	Functional	MH	Low	Previous
7	Shift Management	The system shall allow defining the required roles for each shift.	Functional	MH	Low	Previous
8	Shift Management	The system shall validate HR eligibility.	Functional	MH	Low	Previous
9	Employee Management	The system shall maintain role specific qualifications, including delivery licenses.	Functional	MH	Low	Current
10	Shift Management	The system shall store shift assignments.	Functional	MH	Low	Previous
11	Shift Management	The system shall maintain a history of shifts.	Functional	MH	Low	Previous
12	Shift Management	The system shall provide a user friendly scheduling interface.	Non-Functional	NTH	Low	Future
13	Employee Management	The system shall allow future extension of employee roles.	Functional	NTH	Low	Future
14	Employee Management	The system shall protect employee data from unauthorized access.	Non-Functional	MH	High	Previous
15	Site Management	The system shall allow managing delivery sites.	Functional	MH	Low	Current
16	Site Management	The system shall store each site's address.	Functional	MH	Low	Current
17	Site Management	The system shall store each site's phone number.	Functional	MH	Low	Current
18	Site Management	The system shall store each site's contact person.	Functional	MH	Low	Current
19	Delivery Management	The system shall allow registering deliveries.	Functional	MH	Low	Current

Id	Module	Description	Functional/N on-Functional	Prio rity	Risk	Iteration
20	Delivery Management	The system shall store the source site for each delivery.	Functional	MH	Low	Current
21	Delivery Management	The system shall store the destination site for each delivery.	Functional	MH	Low	Current
22	Delivery Management	The system shall allow assigning a truck to a delivery.	Functional	MH	Low	Current
23	Delivery Management	The system shall allow assigning a driver to a delivery.	Functional	MH	Low	Current
24	Delivery Management	The system shall generate a delivery certificate.	Functional	MH	Low	Current
25	Site Management	The system shall allow managing delivery zones.	Functional	MH	Mid	Current
26	Truck Management	The system shall store each truck's number.	Functional	MH	Low	Current
27	Truck Management	The system shall store each truck's net weight.	Functional	MH	Low	Current
28	Truck Management	The system shall store each truck's maximum carrying weight.	Functional	MH	Low	Current
29	Truck Management	The system shall store the required license type for each truck.	Functional	MH	Low	Current
30	Delivery Management	The system shall calculate the shipment weight.	Functional	MH	Low	Current
31	Delivery Management	The system shall validate the shipment weight against the truck's capacity.	Functional	MH	High	Current
32	Delivery Management	The system shall support delivery replanning when a truck is overloaded.	Functional	MH	High	Current
33	Delivery Management	The system shall record loading confirmation.	Functional	MH	Mid	Current
34	Delivery Management	The system shall maintain the status of each delivery.	Functional	MH	Mid	Current
35	Integration & Validation	The system shall verify that the assigned driver is scheduled to work during the delivery.	Functional	MH	High	Current
36	Integration & Validation	The system shall verify storekeeper availability at the source branch.	Functional	MH	High	Current
37	Integration & Validation	The system shall verify storekeeper availability at the destination branch.	Functional	MH	High	Current
38	Integration & Validation	The system shall validate driver-license compatibility with the assigned truck.	Functional	MH	High	Current

Change Log

Id	Requirement	Change Type	Justification
4	Maintain employee roles and driver license qualifications	Modified	Added driver license qualifications required for transportation.
5	Manage shifts considering transportation planning	Modified	Shift planning must account for transportation staffing needs.
9	Maintain role specific qualifications include delivery licenses	Modified	Extended qualifications to cover delivery licenses.
15	Manage delivery sites	New	Added to manage transportation sources and destinations.
16	Store site address	New	Site data required for deliveries.
17	Store site phone number	New	Site contact data required.
18	Store site contact person	New	Site contact data required.
19	Register deliveries	New	Core transportation capability.
20	Store source site for delivery	New	Required for delivery routing.
21	Store destination site for delivery	New	Required for delivery routing.
22	Assign truck to delivery	New	Required to execute deliveries.
23	Assign driver to delivery	New	Required to execute deliveries.
24	Generate delivery certificate	New	Documentation of each delivery.
25	Manage delivery zones	New	Supports delivery planning by zone.
26	Store truck number	New	Truck identification data.
27	Store truck net weight	New	Needed for capacity calculation.
28	Store truck maximum carrying weight	New	Needed for capacity validation.
29	Store required truck license type	New	Needed for driver-truck compatibility.
30	Calculate shipment weight	New	Needed to validate against truck capacity.
31	Validate shipment weight against capacity	New	Prevents truck overloading.
32	Support delivery replanning when overloaded	New	Handles overload scenarios.
33	Record loading confirmation	New	Tracks the delivery process.
34	Maintain delivery status	New	Tracks the delivery lifecycle.
35	Verify driver scheduled during delivery	New	Integration between shift and delivery scheduling.
36	Verify storekeeper at source branch	New	Loading requires a storekeeper.

Id	Requirement	Change Type	Justification
37	Verify storekeeper at destination branch	New	Unloading requires a storekeeper.
38	Validate driver-license compatibility with truck	New	Ensures legal and safety compliance.

Questions

Employee Module

Topic	Question	Answer
Employee Details	The system is required to store employee details. Should there be an option to update these details? For example, in the case of an employee who changed banks.	Yes, such an option exists
System Login	How will employees log in to the system?	Employees will log in to the system using a username and password
Missing Role	Can a shift be published even when a role is unstaffed?	No, the system requires all roles to be staffed before the next week's shifts are published
Shift Board Update	Can the shift board be updated and changed after publication?	No
Equal Shift Distribution	Should shifts be distributed equally among the different employees?	No
Multiple Branches	Can an employee work in several different branches?	No
Vacation Days	Are there days off during the week or days on which the store is not operating?	At the HR manager's decision
Shift Cancellation	Can an employee cancel a shift after being assigned to it?	No
Bonuses	Are there certain days on which employees receive higher pay than their regular wage?	No
Double Shift	Can an employee work both a morning shift and an evening shift on the same day?	Yes
Shift Submission	Is there a deadline for submitting shifts?	At the HR manager's requirement
Employee Salary	Is there a base salary at which the employee starts working?	At the HR manager's requirement
Access to Employee Details	Who has access to employee details?	Employees (including personal details) and the HR manager (excluding personal details)

Dismissed Employees	Should the system store details of dismissed employees?	Yes
Salary Type	Is the employees' salary hourly or global?	Hourly

Transportation Module

Topic	Question	Answer
Delivery Zones	Can a destination belong to multiple delivery zones?	No, each destination is mapped to a single delivery zone for planning purposes.
Delivery Certificate	Can a delivery certificate be edited after creation?	Yes
Product Weight	Can product weight change over time?	Yes, a product's weight may be updated over time
Truck Usage	Can a truck perform multiple deliveries per day?	Yes
Truck Availability	Can a truck be temporarily unavailable?	Yes, a truck can be temporarily unavailable and the system will not assign it during that period.
Driver Assignments	Can a driver perform multiple deliveries per day?	Yes
Driver Licenses	Can a driver hold multiple license types?	Yes, a driver may hold multiple license types
Overload	Can truck capacity limits be overridden?	No
Source/Destination	Can a branch be both source and destination?	Yes
Delivery Status	Can Delivered status be reverted?	No
Truck Types	Can new truck types be added in the future?	Yes

Integration Questions

Topic	Question	Answer
Driver License	Can an employee hold more than one driving license type?	Yes, a driver may hold more than one license type, the system matches any of them to the truck's required license.
License Ownership	Who updates driver license information?	The HR manager, as part of managing employee details.
License Expiration	Should the system track license expiration dates?	Yes
Driver Assignment	Can a driver be assigned both to a delivery and a regular shift simultaneously?	No
Multiple Deliveries	Can a driver perform multiple deliveries within the same shift?	Yes
Storekeepers	Is a storekeeper required at both source and destination branches?	Yes, a storekeeper is required at both the source (loading) and destination (unloading) branches.
Storekeeper Absence	What happens if a storekeeper becomes unavailable?	The delivery cannot be loaded/unloaded at that branch
Driver Absence	What happens if a driver reports sick after assignment?	The delivery is reassigned to another available, license compatible driver scheduled for that shift and if none exists, the delivery is replanned.
Driving Hours	Is there a maximum number of driving hours per shift?	Yes, bounded by the shift length and legal driving limits.
Working Hours	Are driving hours counted as shift hours?	Yes
Truck Replacement	Can a truck be changed after driver assignment?	Yes
Delivery Cancellation	Should cancelling a delivery automatically update shifts?	No
Permissions	Who can modify driver delivery assignments?	The transportation manager (together with the HR manager).

Assumptions

1. Drivers are employees in the HR system.
2. Every delivery must have exactly one assigned driver.
3. Truck capacity cannot be exceeded.
4. Transportation planning depends on shift assignments.
5. Storekeepers are required for loading and unloading operations.